

Diya Ghosh

West Bengal , India

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🌐 [LinkedIn](#)

🔗 [GitHub](#)

🔗 [LeetCode](#)

👨‍🍳 [Codechef](#)

🌐 [Portfolio](#)

EDUCATION

Indian Institute of Engineering Science and Technology , Shibpur

2023 – 2027(present)

B.Tech-Information Technology- CGPA - 9.37 (till 6th semester)

West Bengal, India

Kamala Girls' School

2021 – 2023

Higher Secondary Education- Percentage - 94.60%

West Bengal, India

EXPERIENCE

Sasken Technologies Limited (Internship)

Dec 2025 – Jan 2026

- Developed a **low-latency voice-to-text system with speaker analytics** during the internship period.
- Integrated the **OpenAI Faster-Whisper (Tiny)** model for **real-time speech transcription**.
- Implemented **emotion, gender analysis** , achieving an end-to-end latency of **under 2 seconds** with structured **JSON** output.

PROJECTS

IT Help Desk Chatbot 🔗 | [Python](#) , [PyTorch](#) , [Streamlit](#)

- Developed an **NLP-based IT support chatbot** that analyzes user queries, classifies them into predefined **intents** and provides **predefined responses**.
- Implemented **text preprocessing techniques** including **tokenization, stemming and Bag-of-Words feature extraction** for accurate intent recognition.
- Built and trained a **neural network model using PyTorch** to improve intent classification performance.
- Created an interactive Streamlit web application with **authentication and chat history management** using **SQLite** .

StudyMate AI 🔗 | [Python](#) , [ChromaDB](#) , [Streamlit](#)

- Developed an **AI-powered learning assistant** that enables users to upload **multiple PDFs** and ask **questions** with context-aware answers including **document sources and page-level citations**.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using **document ingestion, text chunking, embeddings and vector similarity search** for efficient information retrieval.
- Integrated **topic-wise summarization, difficulty-based MCQ generation and interactive answer evaluation** with explanations to enhance learning experience.
- Built and deployed a **Streamlit-based application** with backend integration using **ChromaDB, Sentence Transformers and LLM API**.

Diabetes Predictor 🔗 | [Python](#) , [HTML](#) , [CSS](#) , [JavaScript](#)

- Developed a Diabetes Prediction Application using **Python, FastApi and XGBoost** to predict diabetes risk from user health parameters with **real time confidence scores and health advice**.
- Designed and implemented an interactive web interface using **HTML, CSS and JavaScript** integrated with the FastApi backend for real-time predictions.
- Trained and evaluated **multiple ML models** on the Pima Indians Diabetes Dataset, selecting the best-performing model based on **Accuracy and F1-score** .

TECHNICAL SKILLS

- **Programming Languages** : [Python](#) , [C++](#) , [C](#) , [Java](#)
- **Software Development** : [Object-Oriented Programming \(OOP\)](#) , [Data Structures and Algorithms \(DSA\)](#)
- **Web Development** : [HTML5](#) , [CSS3](#) , [JavaScript](#)
- **Tools , Libraries and Platforms** : [TensorFlow](#) , [PyTorch](#) , [Git](#) , [GitHub](#) , [Pandas](#) , [NumPy](#) , [VS Code](#) , [MySQL](#)

ACHIEVEMENTS

- **UCO Bank Hackathon 2025** 🔗 : – Our **Team Buffer** has secured **3rd place** in the Hackathon organized by UCO bank and IEST Shibpur .
- Secured **3rd position** in the **Hyland Coding Challenge** during a 3-day **Industrial Exposure Program** by **Hyland Software Solutions**. 🔗
- Solved 600+ Problems on **LeetCode** , demonstrating strong DSA algorithmic skills
- **2-star** **Coder** on **CodeChef** with a highest rating of 1521
- **Academic Rank 3** in the department of *Information Technology* till the current semester .